Online Purchasing Behaviour

Project Id: AIML-21

Name : MAYANK MAURYA (Team Leader)

Class Roll Number : 12

University Roll Number : 2028453

Section and Group: G / G1

Name : MOHD UVESH

Class Roll Number: 13

University Roll Number : 2028454

Section and Group: G / G1

1. Problem Description

- In today's digital world, e-commerce platforms generate a vast amount of customer interaction data such as browsing time, clicks, and product views. However, predicting whether a customer will make a purchase remains a challenge. This project aims to build a machine learning model that classifies user behavior into purchasing or non-purchasing categories based on their online activity.
- The system analyses features like session duration, number of clicks, pages visited, and interactions. By applying data preprocessing techniques such as scaling and feature selection, irrelevant or low-impact data is removed to improve performance.
- This project has real-life applications in improving targeted marketing, recommendation systems, and increasing conversion rates for online businesses. It helps companies understand customer intent and optimize their sales strategies effectively.

2. Architecture

1. Data Collection

User-related data is collected from an e-commerce platform dataset.
It includes features such as:

- Browsing time
- Number of clicks
- Product views
- Page values
- Visitor type
- Weekend activity

2. Data Preprocessing

The collected data is cleaned and prepared for model training:

- Handling missing values using mean (numerical) and mode (categorical)
- Removing duplicate records
- Encoding categorical data into numerical form
- Converting boolean values into integers (0/1)
- Feature scaling using standardization

3. Feature Selection

Important features influencing purchase behavior are selected, while irrelevant or low-impact features are removed to improve model performance and reduce complexity.

3. Data Balancing

   Since the dataset may be imbalanced:

- Undersampling is applied
- Equal number of purchase and non-purchase records are maintained

4. Model Training

Machine learning models are trained using the processed dataset:

- Logistic Regression
- K-Nearest Neighbors (KNN)

5. Prediction Module

The trained models predict whether a user will:

- **1 → Make a Purchase (Revenue)**
- **0 → Not Make a Purchase**

6. Model Evalution

The performance of models is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC Score

7. Visualization

Graphs and charts are used to analyze data and model performance:

- Class distribution graph
- Confusion matrix visualization
- Accuracy comparison graph
- ROC curve

3.Models Used(Applied/ Planned) Give a brief introduction of the models used

The following machine learning models are used in this project:

1. Logistic Regression

A classification algorithm used to predict binary outcomes such as pass or fail. It is simple and efficient for linear relationships.

3. K-Nearest Neighbors (KNN)

This algorithm classifies data based on similarity with nearest data points. It works well for smaller datasets.

4.Evaluation metrics used

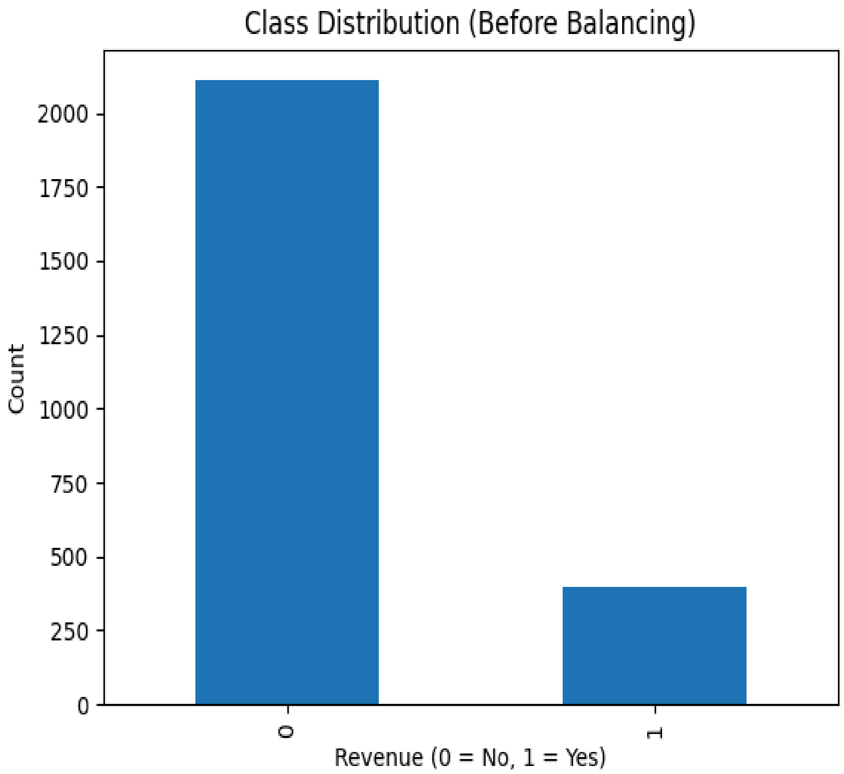
To evaluate the performance of the models, the following metrics are used:

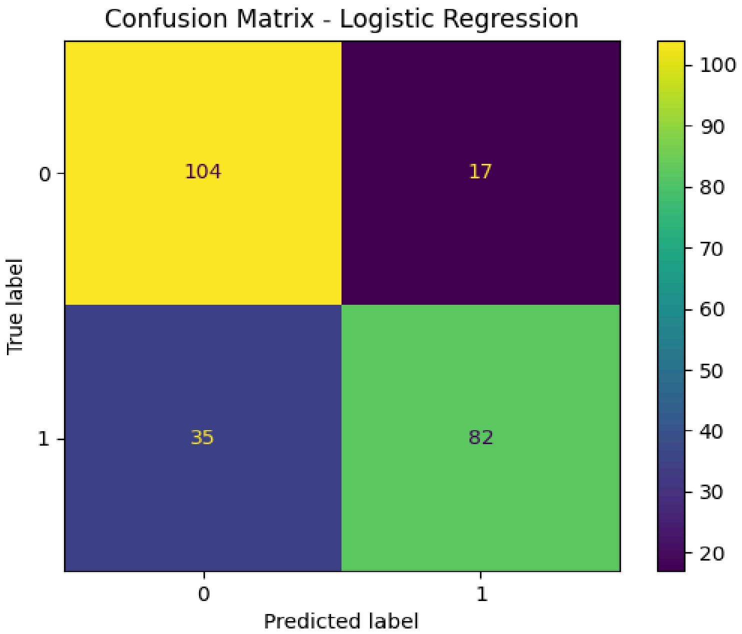
- **Accuracy**

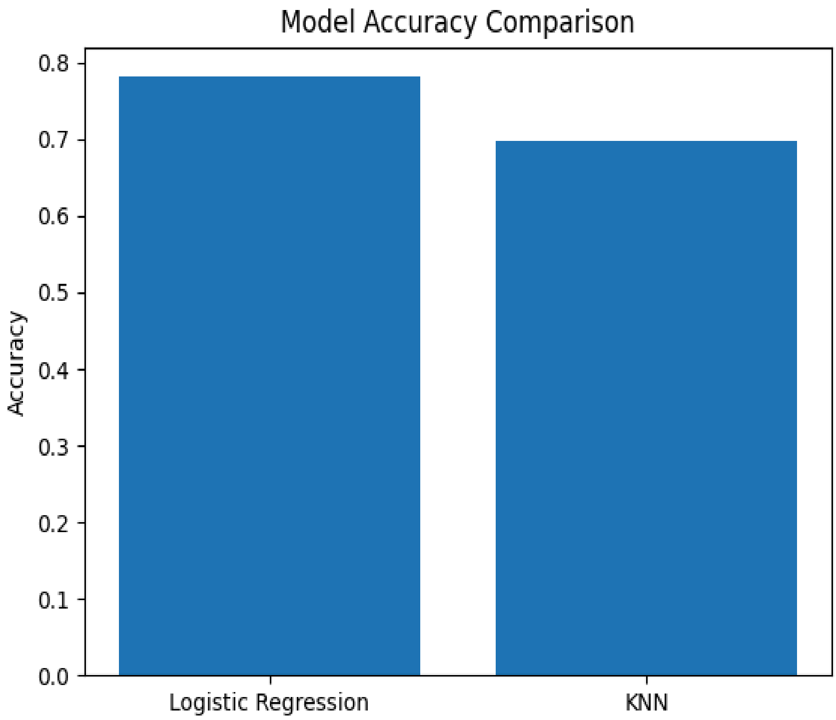
Meas . ures the percentage of correct predictions between (0 to 1) .

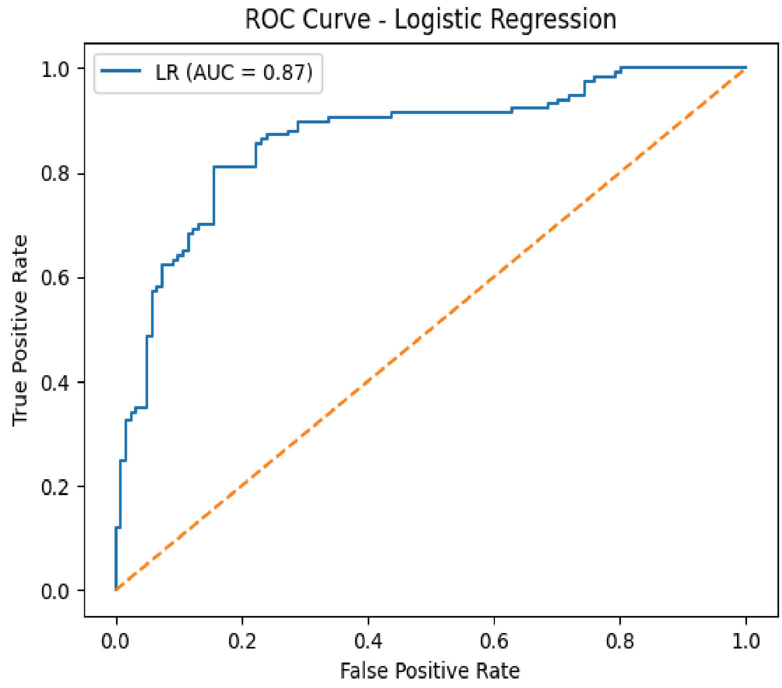
- **Precision**
  Indicates how many predicted positive results are actually correct.
- **Recall**
  Measures how many actual positive cases are correctly identified.
- **F1-Score**
  Harmonic mean of precision and recall, providing a balance between both.
- **Confusion Matrix**
  A table used to visualize model performance.
- **ROC Curve**
  ROC Curve is a graphical representation of classification model at different thresholds.
- **AUC Curve**
  AUC curve represent the area under the ROC Curve

5.Preliminary Results and Visualizations




Confusion Matrix - Logistic Regression
True label
Predicted label
0
1
104
17
35
82
100
90
80
70
60
50
40
30
20


Model Accuracy Comparison
Accuracy
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Logistic Regression
KNN



6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – https://www.geeksforgeeks.org/
- TutorialsPoint – https://www.tutorialspoint.com/
- Stack Overflow – https://stackoverflow.com/
- Kaggle – https://www.kaggle.com/